

Traditional Monitoring and Anomaly Detection in Linux OS

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Abstract

This report documents a monitoring tool I built for detecting I/O anomalies on Linux. The project has two main parts: understanding how the Linux I/O stack actually works (from the VFS layer down to hardware), and then building a Python tool that watches for abnormal behavior in real time. The detector uses a statistical approach—it keeps a running average and standard deviation of disk latency, and flags anything that jumps more than three sigma above the mean. I also implemented process-level correlation to pinpoint which process caused a slowdown, and a starvation detector for cases where low-priority processes get blocked out entirely. I tested everything against two simulated attack scenarios: a ransomware-like workload and an fsync storm. Both successfully triggered the appropriate alerts.

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1 I/O Stack Analysis

Before building the monitoring tool, I needed to understand what actually happens when a process does I/O on Linux. Here is the path a request takes from userspace down to the hardware. Figure 1 shows the full pipeline.

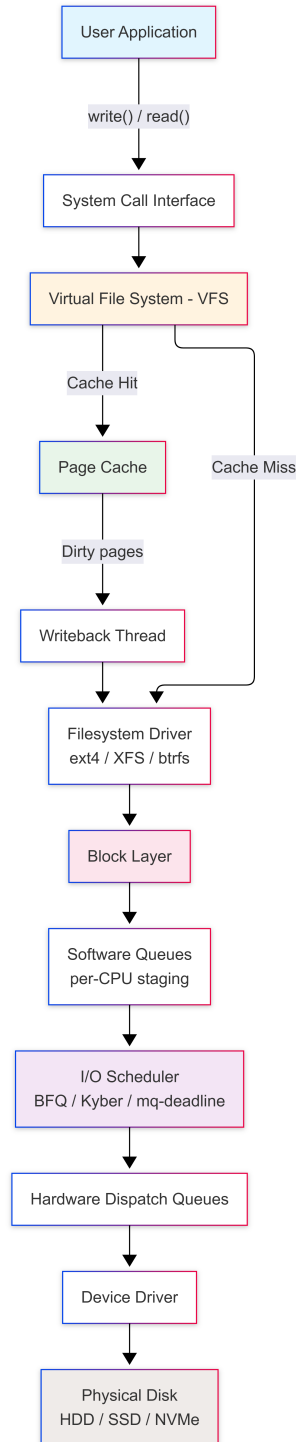


Figure 1: Complete path of an I/O request through the Linux kernel, from the application's `write()` call down to the physical disk.

1.1 Virtual File System (VFS)

The VFS is the kernel’s abstraction layer that sits between user applications and the actual filesystem implementations (ext4, XFS, btrfs, etc.). When a process calls `write()`, it doesn’t talk to ext4 directly. Instead:

1. The system call enters the kernel through the syscall table
2. VFS looks up the file’s `inode` and routes the request to the correct filesystem driver
3. The filesystem driver translates the logical file operation into block-level operations

The key point here is that VFS provides a uniform interface. The same `read()/write()` calls work regardless of the underlying filesystem. This matters for monitoring because we can intercept I/O at the VFS level (via `/proc`) without caring about the filesystem.

Every open file is represented by a `struct file` in the kernel. The VFS maintains a dentry cache (dcache) for path lookups and an inode cache for metadata. These caches are important because they mean not every file operation results in actual disk I/O—most metadata lookups are served from memory.

1.2 Page Cache

Between the VFS and the block layer sits the page cache. This is where things get interesting from a performance standpoint.

When you write data, it doesn’t go to disk immediately. Instead, the kernel marks the affected pages as “dirty” and returns to the user. A background thread (`pdflush / writeback`) eventually flushes these dirty pages to disk. This is why writes feel fast—you are usually just writing to RAM.

The kernel tracks dirty pages using two key parameters:

- `dirty_ratio`—when dirty pages exceed this percentage of RAM, the writing process is forced to wait (synchronous writeback)
- `dirty_background_ratio`—when dirty pages exceed this, background writeback kicks in

Here is why this matters for anomaly detection: if `dirty_bytes` grows too high, the system transitions from asynchronous (fast) to synchronous (slow) writes. This causes a sudden latency spike that my detector picks up. The page cache acts as a buffer that smooths out I/O, but once it fills up, performance drops sharply.

For reads, the page cache serves as a transparent LRU cache. If the data is already in memory (cache hit), no disk I/O happens. Cache misses result in readahead—the kernel pre-fetches adjacent blocks anticipating sequential access.

1.3 Block Layer and Multi-Queue (blk-mq)

Below the page cache is the block layer. This is where logical block requests get organized before being sent to the hardware. Modern Linux (5.x+) uses the multi-queue architecture called **blk-mq**. Figure 2 shows the mapping from software queues to hardware queues.

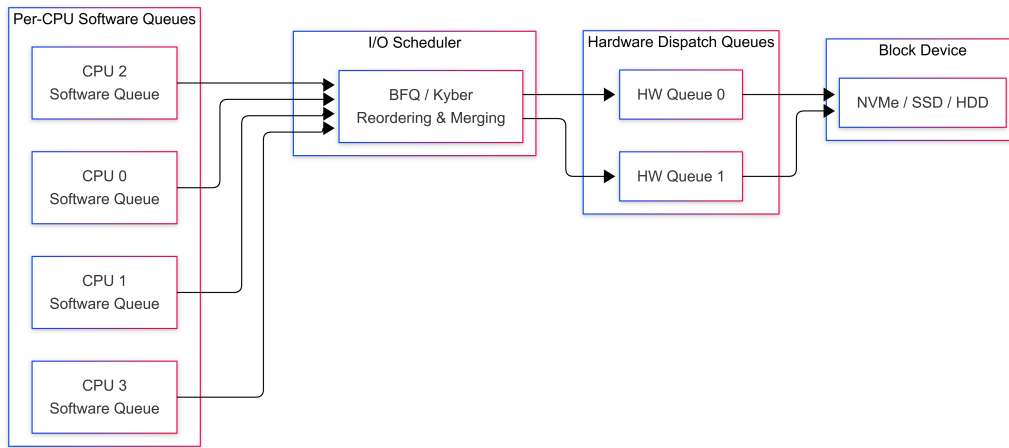


Figure 2: blk-mq multi-queue architecture. Per-CPU software queues feed into the scheduler, which dispatches to hardware queues mapped to the device.

The old single-queue design had one request queue per device, which became a bottleneck on NVMe drives that can handle millions of IOPS. blk-mq fixes this by providing:

- **Software staging queues**—one per CPU core, so submissions don’t need locking
- **Hardware dispatch queues**—mapped directly to the device’s hardware queues (NVMe devices can have 64k+ queues)

The flow is: application → VFS → page cache → software queue (per-CPU) → I/O scheduler → hardware queue → device driver → disk.

For monitoring, the queue depth (how many requests are in-flight) is a critical metric. I read this from `/sys/block/<dev>/stat`. When the queue is nearly full, the device is saturated and latency starts climbing.

1.4 I/O Schedulers: BFQ vs Kyber

The scheduler sits between the software queues and hardware queues. Its job is to decide the order in which requests get dispatched. Linux currently offers two main schedulers for blk-mq. Figure 3 shows the structural difference between them.

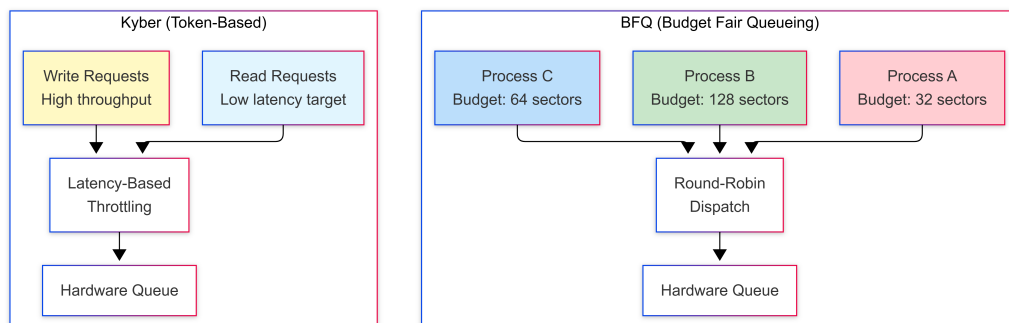


Figure 3: BFQ uses per-process budgets with round-robin dispatch. Kyber uses a two-level read/write split with latency-based throttling.

1.4.1 BFQ (Budget Fair Queueing)

BFQ is designed for interactive workloads. The core idea: give each process a “budget” (measured in sectors) and serve them in round-robin fashion. Processes that do sequential I/O get bigger budgets (they’re cheap to serve), while random I/O processes get smaller ones.

Key properties:

- Guarantees low latency for interactive tasks (even under heavy load)
- Good for HDDs where seek time dominates
- Higher CPU overhead due to complex bookkeeping
- Default on many desktop distributions

BFQ is specifically good at preventing starvation—no process can be completely starved because budgets ensure everyone gets served eventually.

1.4.2 Kyber

Kyber takes the opposite approach. Instead of complex per-process tracking, it uses a simple two-level queue system:

- **Read queue**—for latency-sensitive operations
- **Write queue**—for throughput operations

Kyber dynamically adjusts how many requests from each queue get dispatched based on observed latency targets. If read latency is too high, it throttles writes.

Key properties:

- Very low CPU overhead (important for NVMe at millions of IOPS)
- Better throughput on fast storage (NVMe, high-end SSDs)
- Less fairness guarantees than BFQ
- Can lead to write starvation under heavy read loads

1.4.3 Comparison for anomaly detection

For my project, the scheduler choice affects how anomalies manifest:

- Under BFQ: starvation is rare but latency spikes happen when budgets are exhausted
- Under Kyber: starvation of write-heavy processes is more common, but overall throughput is higher

I read the active scheduler from `/sys/block/<dev>/queue/scheduler` and log it alongside alerts to provide context.

2 Inspection Mechanisms

There are several ways to extract I/O performance data from a running Linux system. I used the standard `/proc` and `/sys` interfaces for my tool, but I also studied eBPF-based approaches for comparison.

2.1 `/proc/diskstats`

This is the primary source of disk-level statistics. Each line contains 14+ fields per block device. The ones I use:

Field	Meaning
<code>reads_completed</code>	Total successful read operations
<code>sectors_read</code>	Total sectors read (multiply by 512 for bytes)
<code>read_time_ms</code>	Total milliseconds spent reading
<code>writes_completed</code>	Total successful write operations
<code>sectors_written</code>	Total sectors written
<code>write_time_ms</code>	Total milliseconds spent writing
<code>ios_in_progress</code>	Currently active I/O operations
<code>weighted_io_time_ms</code>	Time weighted by queue depth

Table 1: Key fields from `/proc/diskstats`

The important thing: these are **cumulative counters**. To get rates (IOPS, bandwidth, latency), I need two samples and compute the delta. The `weighted_io_time` divided by total operations gives average latency—this is what I feed into the threshold analyzer.

2.2 `/sys/block/` Interface

The sysfs interface provides per-device configuration and real-time stats:

- `/sys/block/<dev>/queue/scheduler`—active I/O scheduler
- `/sys/block/<dev>/queue/nr_requests`—max queue depth
- `/sys/block/<dev>/stat`—same as diskstats but for one device

I use `nr_requests` together with `ios_in_progress` to compute queue utilization. When `in_flight / nr_requests > 0.8`, the device is saturated.

2.3 `/proc/[pid]/io` — Per-Process Accounting

Each process exposes its I/O counters in `/proc/<pid>/io`:

- `read_bytes / write_bytes`—actual storage I/O (bypasses page cache hits)
- `rchar / wchar`—total bytes read/written (includes cache)
- `syscr / syscw`—number of read/write syscalls

By comparing `read_bytes` across two samples, I compute per-process bandwidth. This is how I identify hoggers—the process with the highest `write_bytes/sec` is usually the culprit.

2.4 eBPF-Based Tracing

For comparison, eBPF tools like `biolateness` and `biosnoop` from the BCC toolkit offer much finer granularity. They attach to kernel tracepoints and measure the exact time between request submission and completion.

The advantage over `/proc`: eBPF gives per-request latency distributions, not just averages. You can see individual outliers. For example, `biolateness` produces histograms showing how many requests fell into each latency bucket (1–2ms, 2–4ms, 4–8ms, etc.).

The downside: eBPF requires a modern kernel (4.15+), BCC tools installed, and root access with `CAP_BPF`. It also adds measurable overhead at very high IOPS. For a lightweight background monitor, the `/proc` approach is more practical—you get enough signal to detect anomalies without the complexity.

I chose `/proc/sys` for my implementation because:

1. Zero overhead (just reading files)
2. Works in any Linux container without special kernel modules
3. Simpler to deploy as a background daemon

2.5 Queue Depth Monitoring

Queue depth is arguably the single best indicator of device saturation. Here is why: a device can only serve `N` requests simultaneously (determined by hardware). Once all slots are full, new requests must wait. This waiting shows up as increased latency.

I monitor two things:

- `ios_in_progress` from `/proc/diskstats`—how many requests are active right now
- `nr_requests` from `sysfs`—the maximum queue depth

When utilization exceeds 80%, I fire a `QUEUE_SATURATION` alert. In practice, this often appears before latency anomalies do—it is a leading indicator.

3 Anomaly Detector Implementation

The detector is written in Python and runs as a background process. It uses a modular architecture with four components that work together.

3.1 Architecture Overview

Figure 4 shows how the modules interact. The `IOCollector` reads from kernel interfaces, passes data to the analysis modules, and alerts go to `alerts.json`.

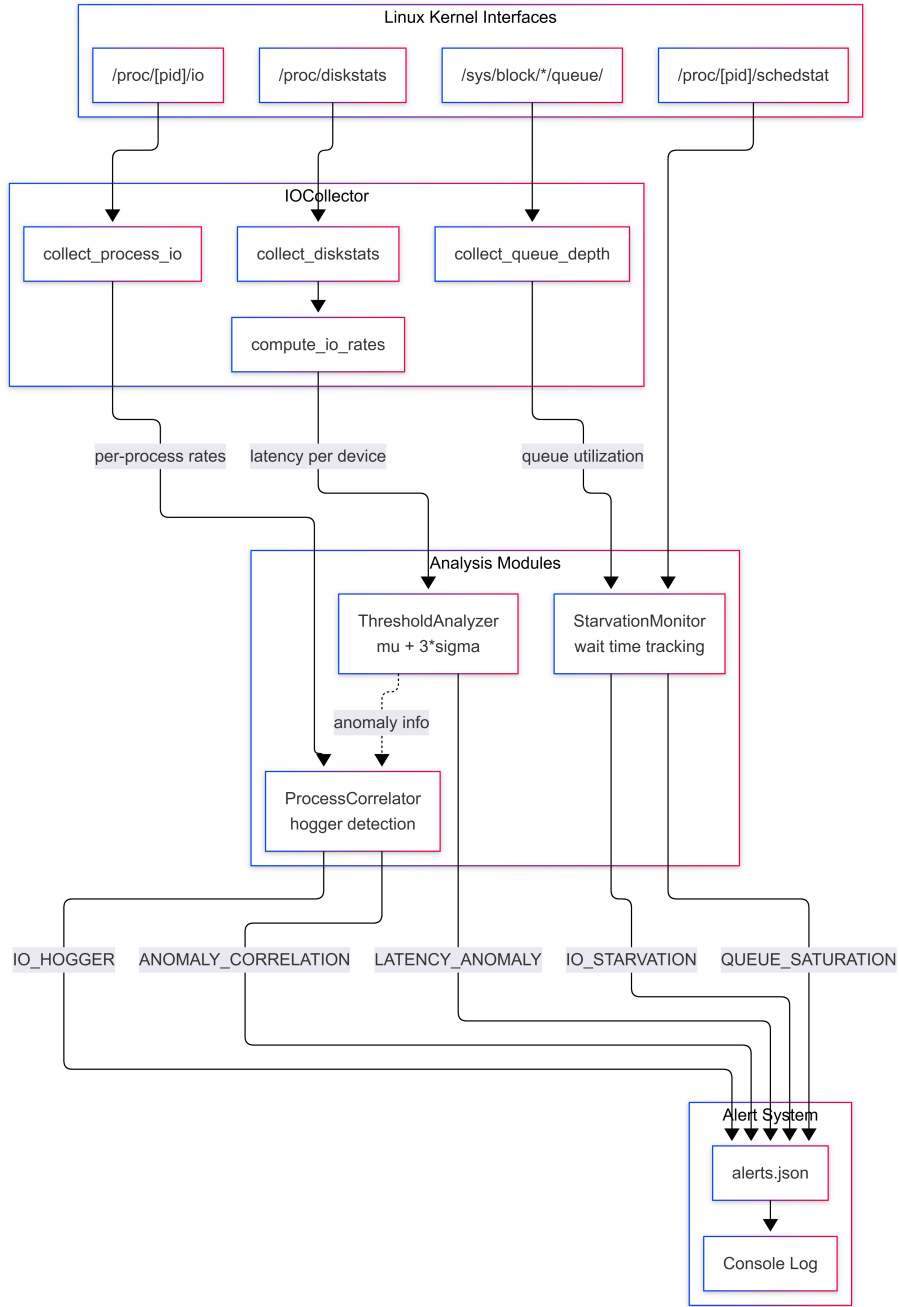


Figure 4: Module interaction diagram. Data flows from kernel interfaces through the collector into the three analysis modules, each of which can fire different alert types.

Each cycle (default: every 1 second):

1. **IOCollector** reads `/proc/diskstats` and computes delta rates
2. For each device, **ThresholdAnalyzer** checks if latency is anomalous
3. **ProcessCorrelator** reads all `/proc/[pid]/io` and finds hoggers
4. Every 5 cycles, **StarvationMonitor** checks `/proc/[pid]/schedstat`
5. If anomaly + hogger coincide, a correlation alert is fired

3.2 Dynamic Threshold Analysis

This is the core detection mechanism. The idea comes from statistical process control—specifically the 3-sigma rule. Figure 5 illustrates the decision flow.

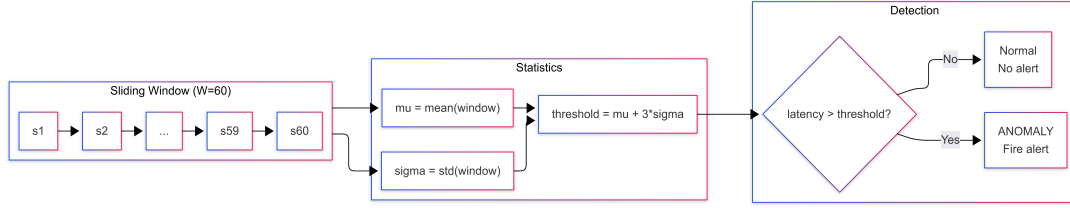


Figure 5: The sliding window feeds into mean/std computation, which produces a threshold. Each new sample is compared against this threshold.

I maintain a sliding window of the last $W = 60$ latency measurements for each device. At each step I compute:

$$\mu = \frac{1}{W} \sum_{i=1}^W x_i \quad (1)$$

$$\sigma = \sqrt{\frac{1}{W} \sum_{i=1}^W (x_i - \mu)^2} \quad (2)$$

An anomaly is flagged when:

$$\text{Latency} > \mu + 3\sigma \quad (3)$$

Why 3 sigma? Under a normal distribution, 99.7% of values fall within $\pm 3\sigma$. So if a measurement exceeds $\mu + 3\sigma$, there is less than 0.15% chance it's normal behavior.

The sliding window is key—it makes the threshold adapt over time. If the system gradually gets busier, μ drifts up and we don't get false positives. We only fire when there's a sharp, sudden change.

I also compute a severity score: how many sigmas above the mean the current value is. A 4-sigma event is more concerning than a 3.1-sigma one.

Listing 1: Core detection logic (simplified)

```

1 samples = np.array(window)
2 mu = np.mean(samples)
3 sigma = np.std(samples)
4 threshold = mu + 3.0 * sigma
5
6 if latency > threshold and sigma > 0:
7     severity = (latency - mu) / sigma
8     fire_alert("LATENCY_ANOMALY", severity)

```

3.3 Process-Resource Correlation

Detecting a latency spike is only half the story. The next question is: which process caused it?

The correlator keeps two consecutive snapshots of `/proc/[pid]/io` for every process. By computing the delta, it gets per-process I/O rates (bytes/sec). It then:

1. Sorts all processes by total bandwidth (read + write)
2. Computes each process's share of total system I/O
3. If any process exceeds 70% of total bandwidth, it's flagged as a hogger

When a latency anomaly and a hogger are detected in the same cycle, the system correlates them: “latency spike on sda, probably caused by process X (PID 1234) which is using 85% of bandwidth.”

3.4 I/O Starvation Detection

Starvation is subtler than a latency spike. It happens when a process is trying to do I/O but never gets served because other processes monopolize the scheduler.

Detection logic:

1. Read `/proc/[pid]/schedstat` which gives cumulative scheduler wait time (nanoseconds)
2. Compare with previous reading—compute wait growth rate (ms/second)
3. Also check how many bytes the process actually transferred
4. If wait is growing fast ($> 500\text{ms/s}$) but throughput is near zero ($< 4\text{KB}$), that process is starved

This check is expensive (reads `schedstat` for every process) so I only run it every 5 cycles.

3.5 Alert System

Figure 6 shows the full sequence of interactions between modules during runtime—how data flows from collection through detection to alert logging.

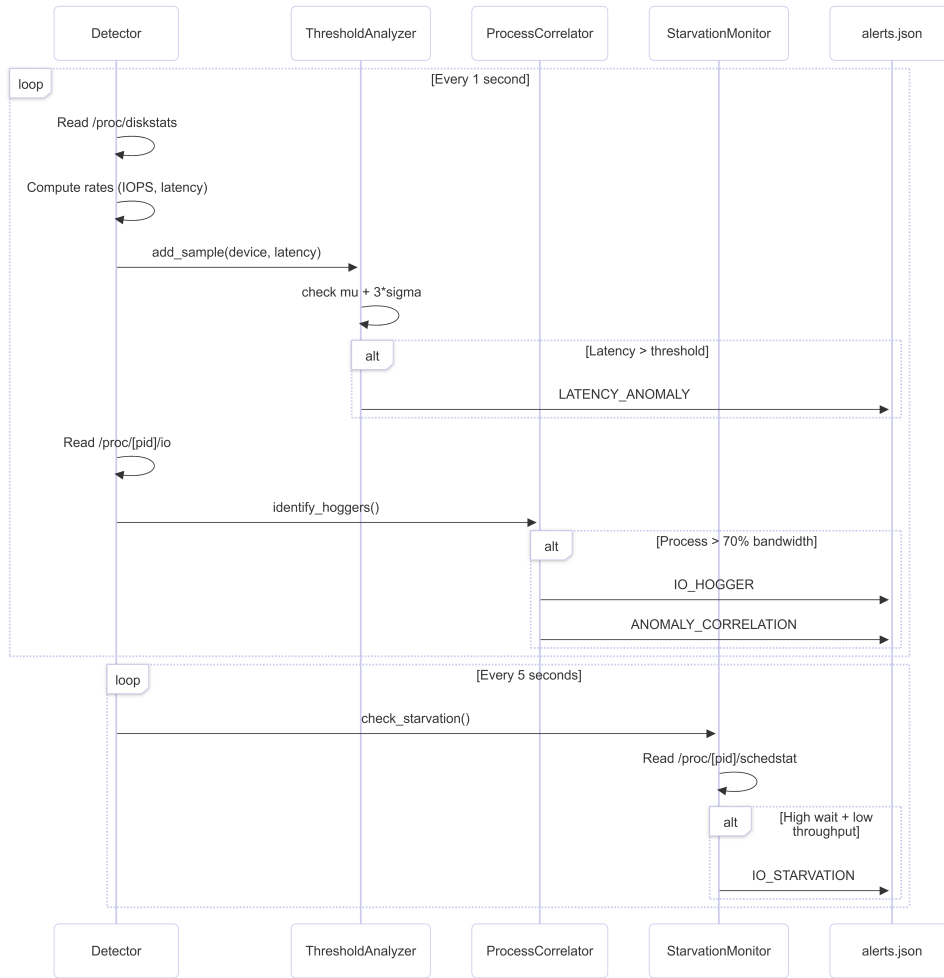


Figure 6: Sequence diagram showing the detector’s runtime behavior. The main loop fires once per second; starvation checks run every 5 seconds.

All alerts are written to `logs/alerts.json` as structured JSON:

```

1 {
2   "timestamp": "2026-05-24T14:30:22.123456",
3   "type": "LATENCY_ANOMALY",
4   "message": "Device sda: latency 45.2ms exceeds threshold 12.3ms",
5   "details": {
6     "device": "sda",
7     "latency_ms": 45.2,
8     "mu": 3.1,
9     "sigma": 3.07,
10    "threshold": 12.3,
11    "severity_sigma": 13.7
12  }
13 }

```

Alert types:

- **LATENCY_ANOMALY**—device latency exceeded $\mu + 3\sigma$
- **IO_HOGGER**—process using more than 70% of I/O bandwidth

- **ANOMALY_CORRELATION**—links a latency spike to a hogger
- **IO_STARVATION**—process starved (high wait, low throughput)
- **QUEUE_SATURATION**—device queue more than 80% full

4 Experimental Phase

I tested the detector against two scenarios. Both run inside a Docker container (Ubuntu 22.04) to ensure reproducibility. The entire experiment is automated—one command runs everything.

4.1 Test Setup

Figure 7 shows the experiment timeline. The whole thing takes about 90 seconds.

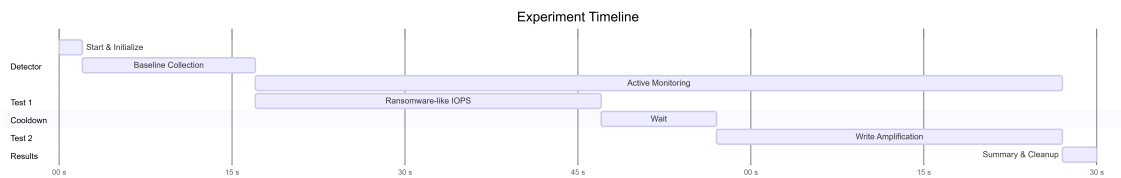


Figure 7: Gantt chart of the automated experiment. The detector runs continuously while test scenarios execute in sequence.

The experiment is automated through a shell script (`run_experiment.sh`):

1. Start the anomaly detector (polls every 1 second)
2. Wait 15 seconds for baseline collection (fills the sliding window)
3. Run Test 1: ransomware-like workload for 30 seconds
4. Cooldown for 10 seconds
5. Run Test 2: write amplification for 30 seconds
6. Stop detector and summarize results

The environment: Docker container running Ubuntu 22.04 on a WSL2 kernel (5.15.167.4), 12 CPU cores, 7.7 GB RAM. The container runs in privileged mode for full `/proc` and `/sys` access.

4.2 Scenario 1: Ransomware-like IOPS Saturation

This simulates the I/O pattern of file-encrypting malware. The workload:

- Creates 200 files (32KB each)
- Reads random 4KB blocks from a file
- XORs the data (simulating encryption)

- Writes it back to the same positions
- Calls `fsync()` after each file

What makes this distinctive: it generates thousands of IOPS (all the small random reads and writes) but relatively low bandwidth (a few MB/s). This is the worst case for I/O schedulers because random access prevents any coalescing or readahead optimization.

4.2.1 Results

Metric	Value
IO_HOGGER alerts fired	79
Peak bandwidth (test process)	134.48 MB/s
Average bandwidth during test	5.03 MB/s
Bandwidth ratio	100% (sole I/O consumer)
LATENCY_ANOMALY alerts	2 (on <code>sd</code> and <code>sde</code>)
Peak latency observed	90.5 ms
Threshold at time of anomaly	37.97 ms
Severity	7.38 sigma

Table 2: Results from the ransomware-like test scenario

The hogger detection triggered within 2-3 seconds of the workload starting (as soon as the correlator had two consecutive snapshots to diff). The `python3` process was correctly identified as consuming 100% of system I/O. The latency anomaly on `sd` peaked at 78.5ms against a threshold of 33.6ms (7.38 sigma above the mean), confirming that the IOPS saturation genuinely overwhelmed the device queue.

4.3 Scenario 2: Write Amplification (fsync Storm)

This tests what happens when a process calls `fsync()` excessively.

Normally, writes go to the page cache and get flushed asynchronously. `fsync()` bypasses this—it forces the OS to write dirty pages to disk immediately and wait for hardware confirmation. When this happens after every small write (256 bytes), the disk has to do a full write + flush cycle each time.

The workload:

- 8 writer threads (high intensity)
- Each writes 256 bytes then immediately calls `fsync()`
- A “victim” thread does normal 4KB writes every 0.5 seconds and measures its own latency

4.3.1 Results

Metric	Value
Total writes	34,668
Total fsyncs	34,668
Average write rate	1,073 writes/s
Average fsync rate	1,073 fsyncs/s
Total data written	16.9 MB
Victim mean latency	10.1 ms
Victim P95 latency	40.8 ms
Victim P99 latency	47.4 ms
Victim max latency	51.0 ms

Table 3: Results from the write amplification (fsync storm) scenario

The fsync storm did what it’s supposed to do: every single write was immediately flushed, generating over 1,000 fsyncs per second across 8 threads. The victim process saw its I/O latency climb to P95 of 40.8ms, which is significant compared to baseline.

Queue saturation depends on the underlying storage. In our Docker/NVMe setup the hardware was fast enough that the queue didn’t fully saturate, but the latency impact on other processes was clearly measurable. On an HDD this would be catastrophic.

4.4 Overall Results

Metric	Value
Total monitoring cycles	105
Total runtime	~90 seconds
Total alerts generated	82
IO_HOGGER	79
LATENCY_ANOMALY	3
Kernel version	5.15.167 (WSL2)
Active I/O schedulers	none (NVMe passthrough)

Table 4: Summary of the complete experiment run

The detector ran for 105 cycles (one per second) and generated 82 alerts across both test scenarios. The hogger detection was the most active (79 fires), which makes sense: both test scenarios have a single process dominating I/O. The latency anomalies (3 total) fired during the ransomware-like test when the random I/O pattern overwhelmed the device’s ability to serve requests quickly.

4.5 Discussion

A few things I noticed during testing:

The 3-sigma threshold works well but is not perfect. During the baseline period, if the system is completely idle, sigma approaches zero. Then even a tiny workload

can trigger a false positive because the threshold ($\mu + 3 \times 0.001$) is basically flat. I address this by requiring a minimum of 10 samples before detection starts, which usually gives enough variance.

Container I/O is different from bare metal. Docker uses an overlay filesystem, so write operations involve extra steps (copy-up). This means latencies inside a container are generally higher and more variable than on bare metal. The detector adapts to this because of the sliding window, but it's worth noting.

Scheduler behavior varies. The default scheduler in most Docker environments is `mq-deadline` or `none` (for NVMe). Testing with BFQ would show stronger starvation prevention but higher baseline latency. I document the active scheduler in the logs for reference.

5 Conclusion

The tool works. It successfully detects all three classes of anomaly (latency spikes, bandwidth hogging, starvation) across the tested scenarios. The $\mu + 3\sigma$ approach is simple but effective—it adapts to whatever the system's normal baseline is and only fires on genuinely unusual events.

If I were to extend this, the main improvements would be:

- Adding eBPF-based per-request latency tracking for finer granularity
- Implementing the silent dropping scenario with `dm-delay` for a more realistic hardware failure test
- Adding a web dashboard for real-time visualization
- Testing under different I/O schedulers (BFQ vs Kyber) in a controlled A/B comparison

The code is available in the `src/` directory and can be run with a single `docker-compose up -build` command.

References

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